

Dolutegravir in HIV-2–Infected Patients With Resistant Virus to First-line Integrase Inhibitors From the French Named Patient Program

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Background. Dolutegravir has shown in vitro activity against human immunodeficiency virus type 2 (HIV-2). We report safety and efficacy data of regimens containing dolutegravir (50 mg twice daily) in antiretroviral-experienced, HIV-2–infected patients.

Methods. HIV-2–infected patients experiencing virological failure to raltegravir received dolutegravir with optimized background antiretroviral combinations within the French Named Patient Program (NPP). Plasma HIV-2 RNA (pVL) was assessed at time of dolutegravir initiation (baseline), month 3, and month 6. Antiretroviral trough plasma concentrations (C_{12h}) were determined using liquid chromatography coupled with tandem mass spectrometry.

Results. Thirteen HIV-2–infected-patients, with a median duration of 15 years' infection and given 16 previous antiretroviral regimens, were included in NPP. Median follow-up was 9 months (min–max, 3–15 months). Median baseline pVL and CD4 cell count were 9544 copies/mL (inter quartile range [IQR], 3096–23 120 copies/mL) and 100 cells/ μ L (IQR, 77–171 cells/ μ L), respectively. Available integrase genotypic resistance patterns were Y143C/G/H/R (n = 5), Q148R/K (n = 2), and N155H (n = 4). Optimized background antiretroviral regimens conferring a genotypic sensitivity score ≤ 2 in 10 patients included nucleoside reverse transcriptase inhibitors associated with darunavir/ritonavir (n = 12), saquinavir/ritonavir (n = 2), and maraviroc (n = 3). At months 3 and 6, pVL was undetectable in 6 of 13 and 4 of 12 patients, respectively, and median CD4 count was 161 (101–188) cells/ μ L and 167 (135–1353) cells/ μ L, respectively. Median dolutegravir C_{12h} was 4086 (1756–5717 ng/mL) ng/mL in 9 patients. No serious events were notified except 1 death from progressive multifocal leukoencephalopathy at month 4.

Conclusions. Optimized dolutegravir-containing antiretroviral regimens supported by good plasma exposure provide a substantial initial efficacy rate for salvage therapy in heavily antiretroviral-experienced HIV-2–infected patients with virus harboring resistance to first-generation integrase inhibitors. Larger numbers of patients and longer follow-up are needed to confirm these findings.

Keywords. HIV-2; dolutegravir; integrase; drug resistance mutations.

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Although human immunodeficiency virus type 1 (HIV-1) is responsible for the AIDS pandemic, approximately 1–2 million people worldwide are infected with human immunodeficiency virus type 2 (HIV-2), most of them in Senegal, Guinea Bissau, Gambia, Ivory Coast, and Cape Verde [1]. In addition to being endemic in West Africa, HIV-2 has spread to 19 different countries in

Europe, Asia, and North America [2–6]. In Europe, a relatively high prevalence of HIV-2 infection has been reported in Portugal [6] and France, where surveillance studies showed that about 2% of new infections in 2003–2006 were caused by HIV-2 [2]. In 2007, raltegravir, the first integrase strand transfer inhibitor (INSTI), was approved by the US Food and Drug Administration (FDA) after studies showed it to be efficacious in both antiretroviral (ARV)–naïve and -experienced HIV-1–infected participants when dosed twice daily [7]. The next INSTI, elvitegravir, was approved in 2012 as part of a fixed-dose daily tablet containing tenofovir disoproxil fumarate, emtricitabine, and the cytochrome P450 isoenzyme 3A inhibitor cobicistat [8, 9]. However, elvitegravir is only available in fixed-dose combination for the treatment of ARV-naïve HIV-1–infected patients and requires pharmacologic boosting to be given once daily. In addition, significant cross-resistance between raltegravir and elvitegravir prevents sequential therapy with these 2 drugs [10, 11]. Recently, the FDA approved dolutegravir as the newest INSTI for treatment of HIV-1 in ARV-naïve and -experienced persons based on studies demonstrating efficacy and safety. The drug has advantages over prior INSTI in that it can be given once daily without boosting and can overcome some prior INSTI failures.

In France, dolutegravir recently received approval by Agence Nationale de Sécurité du Médicament et des Produits de Santé in 2014. Since December 2011, dolutegravir has been available under the dolutegravir French Named Patients Program (NPP) for salvage therapy in patients infected with INSTI-resistant HIV-1 and HIV-2 and experiencing treatment failure without any treatment options.

Despite 40% difference in amino acid sequence between HIV-1 and HIV-2 integrases, phenotypic assays performed with reference strains or clinical isolates have shown that all approved integrase inhibitors are effective against HIV-2 [12–15]. There is limited information on the mutational pathways leading to HIV-2 resistance to approved integrase inhibitors. However, the available evidence indicates that mutations conferring resistance in HIV-1 and HIV-2 are essentially the same [12, 16]. In both viruses, raltegravir and elvitegravir have a low to moderate genetic barrier and show extensive cross-resistance. On the other hand, dolutegravir exhibits a predictable pharmacokinetic profile and a well-defined exposure–response relationship [17] and retains activity against raltegravir- and elvitegravir-resistant HIV-1 strains [18, 19], but the information on its efficiency against HIV-2 strains bearing integrase inhibitor resistance mutations is still limited [12]. Here, we report preliminary safety and efficacy and pharmacokinetic data of regimens containing dolutegravir (50 mg twice daily) in HIV-2–infected patients with virus harboring INSTI resistance–associated mutations from the French NPP.

METHODS

Patients

HIV-2–infected adults (≥ 18 years of age) experiencing virological failure to raltegravir-containing regimens, with virus harboring INSTI resistance–associated mutations (at codons 66, 92, 97, 138, 140, 143, 147, 148, or 155) received dolutegravir 50 mg open label twice daily with optimized background ARV combinations, based on baseline genotypic resistance tests and/or previous treatment failures. The program was conducted between December 2011 and September 2013 and was approved by the French regulatory authority and by the ethics review committee. Written informed consent was obtained from each individual before screening. Demographic, biological, and therapeutic characteristics were recorded.

Virological Analyses

Plasma HIV-2 RNA quantification was performed by using an in-house real-time reverse transcription polymerase chain reaction with a limit of detection of 40 copies/mL [20]. Samples were collected at time of dolutegravir initiation (baseline), month 3, and month 6. Direct sequencing of protease, reverse transcriptase, and integrase genes was performed as previously described [14, 21]. Viral tropism was genotypically predicted as previously described [22]. Genotypic resistance tests were interpreted according to the Agence nationale de recherche sur le sida et les hépatites virales algorithm, version 23 (www.frenchresistance.org). Genotypic sensitivity score (GSS), taking into account foscarnet susceptibility, was calculated for each patient. HIV-2 group was evaluated by phylogenetic analyses of the reverse transcriptase sequences and the consensus HIV-2 sequences using the Los Alamos database (www.hiv.lanl.gov) as previously described [23, 24].

Pharmacokinetic Analyses

Blood samples were collected during follow-up 12 ± 2 hours after the last drug intake (C_{12h}). Simultaneous ARV plasma determinations were performed using liquid chromatography coupled with tandem mass spectrometry (Acquity UPLC/TQD, Waters Corporation, Milford, Massachusetts) according to a modified previous method [25]. Considering the lack of in vivo data on HIV-2 threshold plasma concentrations and according to the respective daily dosage administered (50 mg twice daily) to patients infected with HIV-2 harboring INSTI resistance–associated mutations experiencing virological failure to raltegravir-containing regimen, C_{12h} was considered as adequate if higher than the respective minimal value at the considered daily dose: dolutegravir ≥ 1000 ng/mL, darunavir ≥ 2000 ng/mL, and maraviroc ≥ 50 ng/mL [26–28].

RESULTS

A total of 13 ARV-experienced HIV-2–infected patients (10 men and 3 women) with a median age of 51 years (interquartile

Table 1. Patient Characteristics

Patient	Age, y	Sex	Duration of Known Infection, y	DTG-Containing Antiretroviral Regimens	Viral Tropism	Baseline		Month 3		Month 6	
						Plasma RNA, Copies/mL	CD4 Count, Cells/ μ L	Plasma RNA, Copies/mL	CD4 Count, Cells/ μ L	Plasma RNA, Copies/mL	CD4 Count, Cells/ μ L
1	53	M	9	ZDV/3TC/DRV/r/PFA	X4	46 760	439	456	324	3170	450
2	43	M	15	TDF/FTC/LPV/SQV/r	X4	23 120	85	<40	181	<40	NA
3	52	M	9	ZDV/ABC/3TC/DRV/SQV/r/PFA	X4	1280	100	<40	132	<40	140
4	48	M	22	ZDV/3TC/DRV/r/PFA	X4	3096	53	376	76	807	46
5	53	M	13	DRV/r/MVC/PFA	R5	26 060	157	<40	157	<40	320
6	54	M	10	3TC/DRV/r/MVC/PFA	R5	138 400	62	11 440	45	9991	174
7	69	M	20	ZDV/3TC/DRV/r/MVC	NA	9544	138	1119	168	6715	118
8	51	F	20	DRV/SQV/r	R5	3050	171	<40	157	<40	159
9	50	F	2	DRV/r/MVC	X4	18 630	355	<40	342	323	616
10	33	M	11	ABC/DRV/r	X4	861	77	<40	109	NA	NA
11	58	M	24	TDF/ddl/DRV/r	X4	3998	27	5407	NA	4795	NA
12	64	M	15	ABC/3TC/DRV/r	X4	21 199	97	1450	34	20 206	NA
13	51	F	22	TDF/FTC/DRV/r	X4	5741	200	6083	207	3995	NA

Abbreviations: 3TC, lamivudine; ABC, abacavir; ddl, didanosine; DRV/r, ritonavir-boosted darunavir; DTG, dolutegravir; FTC, emtricitabine; LPV, lopinavir; MVC, maraviroc; NA, not available; PFA, foscarnet; SQV/r, ritonavir-boosted saquinavir; TDF, tenofovir disoproxil fumarate; ZDV, zidovudine.

range [IQR], 50–54 years) received dolutegravir in the NPP. These patients received a median of 11 previous ARV drugs (min–max, 6–17) before the dolutegravir-containing regimen. The characteristics of the patients at inclusion are summarized in the Tables 1 and 2. The median duration of HIV-2 known infection was 15 years (IQR, 10–20 years). The median follow-up under dolutegravir was 9 months (range, 3–15 years). HIV-2 groups were categorized as A (n = 9), B (n = 3), and H (n = 1).

At baseline, median plasma HIV-2 RNA level and CD4 cell count were 9544 copies/mL (3096–23 120 copies/mL) and 100 cells/ μ L (77–171 cells/ μ L), respectively. HIV-2 tropism was genotypically predicted as CCR5 in 3 patients, whose CD4 counts were 62, 157, and 171 cells/ μ L, respectively. Optimized background ARV regimens included nucleoside reverse transcriptase inhibitors (tenofovir disoproxil fumarate [n = 3], zidovudine [n = 4], lamivudine or emtricitabine [n = 8], abacavir [n = 3], didanosine [n = 1]) associated with ritonavir-boosted darunavir 600/100 mg twice daily (n = 12), ritonavir-boosted saquinavir 1000/100 mg twice daily (n = 3), and maraviroc 150 mg twice daily (n = 3). Dual boosted protease inhibitor regimens consisting of ritonavir-boosted darunavir/saquinavir and ritonavir-boosted lopinavir/saquinavir were administered to 3 patients (patients 2, 3, and 8). Five patients also received a foscarnet 3-week induction treatment associated with zidovudine in 4 cases. Genotypic sensitivity score was 3 (n = 3), 2 (n = 6), and 1 (n = 4), respectively. Baseline patterns of protease,

reverse transcriptase, and integrase resistance-associated mutations and tropism are depicted in Table 2. Available integrase genotypic resistance patterns were Y143C/G/H/R (n = 5), Q148K/R/S (n = 2), and N155H (n = 4). Among them, T97A was present in all Y143 patterns. The E157Q mutation was observed in 2 cases, 1 with the Y143C/G/H/R and 1 with the N155H integrase-associated resistance mutations. Baseline integrase resistance pattern was missing in 1 patient due to lack of gene amplification. Table 2 depicts baseline patterns of integrase-associated mutations and corresponding baseline GSS, plasma HIV-2 RNA decrease between baseline and month 6, and median dolutegravir C_{12h} for each patient receiving the dolutegravir-containing regimen. At month 3, plasma viral load below the limit of detection was achieved in 5 of the 13 patients. Viral load remained undetectable in the 4 patients tested at month 6. Median decrease in plasma viral load was -1.31 (IQR, -2.01 to -0.93) $\log_{10}$ copies/mL and -1.16 (IQR, -1.79 to -0.16) $\log_{10}$ copies/mL at months 3 and 6, respectively. Median CD4 count rose from 100 cells/ μ L (IQR, 77–171 cells/ μ L) to 158 cells/ μ L (IQR, 101–188 cells/ μ L) and to 166 cells/ μ L (IQR, 135–353 cells/ μ L) at month 3 and month 6, respectively. Among the 5 patients receiving foscarnet induction treatment, only 2 had undetectable plasma viremia at months 3 and 6. All patients with undetectable plasma HIV-2 RNA at month 6 had virus harboring the integrase resistance pattern Y143C/G/H/R at baseline. Median dolutegravir C_{12h} was 4086 ng/mL (IQR, 1756–5717 ng/mL) in

Table 2. HIV-2 Plasma Viremia Decrease From Baseline at Month 6 According to Baseline Resistance Patterns, Baseline Genotypic Sensitivity Score, and Median Dolutegravir Plasma Concentration 12 Hours After the Last Drug Intake Determined During Follow-up

Patient	Baseline Genotypic Sensitivity Score	Baseline Resistance-Associated Mutations			Delta (M6-BL) Plasma RNA, log ₁₀ copies/mL	Median DTG C _{12h} , ng/mL (No.)
		Protease Gene	Reverse Transcriptase Gene	Integrase Gene		
1	3	10I, 46L, 47A, 71I, 84V, 89V, 90M	65R	155H	−1.77	2708 (4)
2	2	53L, 54L, 71I	NA	97A, 143C/G	−2.77	6039 (1)
3	3	10I, 47A, 71I, 84V, 90M	184V	97A, 143H/R, 157Q	−1.52	4801 (3)
4	1	54M, 71I, 82F, 90M	65R, 70R, 151M	72I, 155H	−0.58	4360 (2)
5	2	10I, 50V, 71I, 82F, 89V, 90M	67N, 115F, 151M, 184V	97A, 143G	−2.82	NA
6	2	54M, 71I, 82F, 89V, 90M	65R, 67M, 151M, 184V	92Q, 155H, 157Q	−1.14	1367 (6)
7	2	54M, 71I, 84V, 90M	65R, 69S, 184V,	140S, 148R	−0.15	2453 (1)
8	1	50V, 54L, 71I, 90M	65R, 151M, 184V	72I, 97A, 143R	−1.89	5496 (5)
9	1	10I, 50V, 54M, 71I	65R, 70R, 115F, 184V	92Q, 155H	−1.76	4717 (2)
10	2	71I	184V	72I, 97A, 143R	NA	NA
11	3	54M, 71I, 82F, 90M	67N, 184V, 215Y	147G	0.08	5159 (2)
12	2	10I, 50V, 89V	184V, 215Y	NA	−0.02	NA
13	1	15V, 54M	65R, 151M, 184V	140S, 148K	−0.16	NA

Abbreviations: C_{12h}, median plasma concentration 12 hours after last drug intake; DTG, dolutegravir; NA, not available.

9 patients during follow-up (Table 2). Median darunavir C_{12h}, ritonavir C_{12h}, and maraviroc C_{12h} were 2981 ng/mL (IQR, 2169–5082 ng/mL), 250 ng/mL (IQR, 148–495 ng/mL), and 219 ng/mL (IQR, 219–555 ng/mL), respectively. Eighty-eight percent of dolutegravir C_{12h} (23/26 measurements in 9 patients), 79% of darunavir C_{12h} (19/25 measurements in 9 patients), and 100% of maraviroc C_{12h} (11/11 measurements in 3 patients) were considered as adequate during the follow-up. At month 6, despite the small number of samples, median dolutegravir C_{12h} in patients with plasma HIV-2 RNA <40 copies/mL (5496 ng/mL) tended to be higher than those in patients with detectable plasma HIV-2 RNA at month 6 (2687 ng/mL). Moreover, among the 5 patients demonstrating a viral load drop of at least −1.5 log₁₀ copies/mL between baseline and month 6, median dolutegravir C_{12h} was >4000 ng/mL in the 4 patients with available data. The last patient had no sample available for therapeutic drug monitoring.

One patient (patient 2) died from multifocal leukoencephalopathy at month 4. No other serious clinical or biological events led to dolutegravir discontinuation.

At month 6, integrase genotypes were available in 7 patients. The selection of the T97A mutation was observed in 2 patients (patients 1 and 4). No change in the integrase gene could be evidenced in the 5 remaining patients.

At month 12, plasma viremia was available in 9 of the 13 patients. Plasma viremia was undetectable in 3 patients (patients 3, 6, and 8) and close to month 6 values in the 6 remaining patients (patients 1, 4, 7, 11, 12, and 13).

DISCUSSION

This study is the first to explore efficacy and safety of a dolutegravir-containing salvage regimen in HIV-2–infected patients who had experienced virological failure while receiving raltegravir-containing regimens and had genotypic evidence of raltegravir resistance. Despite the small number of patients, these preliminary results demonstrate that dolutegravir-containing regimens provide a substantial efficacy rate for salvage therapy in heavily ARV-experienced patients with virus harboring resistance to first-generation INSTIs. Sustained virologic response up to 6 months was observed in one-third of the patients, although immunological recovery varied depending on the individual. In all of these patients, plasma exposures of darunavir and maraviroc were adequate and particularly elevated for dolutegravir, demonstrating good treatment adherence and probably optimized for highly resistant virus.

Dolutegravir was well tolerated when administered at 50 mg twice daily, consistent with safety data from other treatment studies [29, 30], although the sample size was small. One patient died from progressive multifocal leukoencephalopathy at month 4 despite a good virological response at month 3.

As with other ARV drugs, durability of response is more likely when there is support from background drug activity [31]. In our study dealing with multiexperienced HIV-2–infected patients, at least 2 active drugs in combination with dolutegravir were available for 9 patients. However, cohort studies of HIV-2–infected patients are limited and have not addressed the central

questions regarding the optimal therapy for this virus as no large clinical trials are available. Moreover, HIV-2 genotypic resistance interpretation systems, relying on small studies, are most of the time based on genotype/phenotype correlations and not on virological response as for HIV-1. Therefore, GSS calculated for HIV-1 might be overestimated for HIV-2. This possible weakness regarding genotyping interpretation might explain why all patients did not respond to dolutegravir regimens despite a GSS ≥ 2 .

Foscarnet, a pyrophosphate analogue, prevents pyrophosphate exchange and thereby inhibits DNA chain elongation catalysed by a variety of viral DNA polymerases including HIV reverse transcriptase. Foscarnet associated or not with zidovudine has been proposed as salvage therapy for multi-experienced patients with late-stage HIV-1 infection and limited therapeutic options [32, 33]. Thymidine-associated mutations appear to render HIV hypersusceptible to foscarnet in vitro. In HIV-2 infection, 1 case report described a marked decrease of plasma HIV-2 load and improved immunological status in a heavily ARV-experienced HIV-2-infected patient exhibiting multidrug-resistant virus [34]. Recently, a small series of HIV-1-infected patients reported efficacy of the addition of foscarnet and zidovudine to various dolutegravir combinations in salvage therapy [35]. In our study, 5 patients received foscarnet associated with zidovudine during an induction period of 3 weeks, and some experienced favorable response to dolutegravir combination therapy. Indeed, 2 of them had an undetectable plasma viral load at months 3 and 6. Viral load drop at month 3 varying between -0.92 and $-2.82 \log_{10}$ copies/mL was maintained above $-1.0 \log_{10}$ copies/mL in 4 of the 5 patients at month 6.

In HIV-1, in vitro data as well as clinical data indicate that primary mutations at codon 155 or 143 and the T66I and E92Q mutants remain susceptible to dolutegravir, whereas mutations at codon 148 in the presence of other secondary mutations can lead to decreased dolutegravir efficacy [36, 37]. Indeed, in a subanalysis of cohort II of the VIKING study, participants with current or previous raltegravir failure and triple-class resistance (including INSTI) received dolutegravir 50 mg twice daily in addition to their failing regimen for 10 days, then dolutegravir plus an optimized background regimen. Twelve of 13 participants with baseline Y143 and N155 mutations achieved HIV-1 RNA load <400 copies/mL at week 24, whereas 8 of 11 participants with Q148 plus ≥ 1 additional raltegravir-associated mutation achieved a viral load <400 copies/mL at week 24 [37]. In the VIKING-3 trial, 183 participants with raltegravir and/or elvitegravir-resistant virus plus resistance to at least 2 other ARV drug classes received open-label dolutegravir 50 mg twice daily while continuing their failing regimen (without raltegravir or elvitegravir). At day 8, an optimized background regimen was started that included dolutegravir. Participants with the T66, Y143, or N155 mutations had

antiviral efficacy at week 24 of 100%, 75%, and 88%, respectively, indicating these that INSTI mutations alone have little effect on dolutegravir. Those with a Q148 plus 1 secondary mutation or Q148 plus at least 2 secondary mutations had 59% and 24% response rates, respectively, indicating that the Q148 mutation pathway can lead to decreased efficacy of dolutegravir as the number of additional mutations increases [29].

In our study, all patients with codons 148 or 155 integrase inhibitor resistance-associated mutations achieved a detectable plasma viremia at month 6, indicating that these profiles impact dolutegravir activity in HIV-2. These data are consistent with in vitro susceptibility findings obtained on clinical isolates or site-directed mutants [12, 38]. However, the impact of the Y143C mutation on dolutegravir susceptibility is less clear. In our study, the 4 evaluable patients with virus harboring this integrase mutation resistance profile, in association with 72I, 97A, and 157Q mutations and without 148 or 155 mutations, achieved undetectable viral load at months 3 and 6, showing that, as in HIV-1, this mutation may have no or little effect on dolutegravir susceptibility in vivo. However, in vitro the Y143C mutation in HIV-2 has been associated with reduced susceptibility to dolutegravir using different types of phenotypic assays on clinical isolates and recently on site-directed mutants [12, 38]. Although, in our series, the 143C amino acid substitution was present in only 1 case in mixture with 143G, the other mutants involving 143H/R or 143R in 3 cases and 143G in 1 case. This indicates that the different substitutions selected at codon 143 may not impact dolutegravir activity in the same way, Y143C being more deleterious. This has been recently described by Huang et al in HIV-1 regarding the codon 148 where the Q148K substitution displayed larger reductions in dolutegravir susceptibility than corresponding Q148H/R [39, 40]. One patient with virus harboring the 147G integrase resistance mutation, previously described as associated with high level of resistance to elvitegravir in HIV-1, did not respond to dolutegravir, suggesting that this mutation might impact dolutegravir activity [41]. In vitro phenotypic susceptibility assay of specific mutant generated by site-directed mutagenesis is warranted to confirm these data.

In our series, the R263K substitution, described in HIV-1 as polymorphism but also selected under dolutegravir selective pressure [18, 42], was not present at baseline in the 7 patients with detectable plasma viremia at month 6 for whom genotyping was available. The selection of additional T97A integrase resistance mutation was evidenced at month 6 in 2 patients harboring virus with N155H resistance profile, indicating that this mutation might play a role in dolutegravir resistance. Larger studies are needed to confirm these data.

In our series, the plasma exposure of ritonavir-boosted darunavir and maraviroc was adequate in the majority of the patients and that of dolutegravir was particularly elevated, partly

because of double daily dose and partly because of food effect, demonstrating good treatment adherence during the follow-up despite the pill burden of twice-daily dosing in association with foscarnet intensification. These plasma exposures were also particularly appropriate and optimized for highly resistant HIV-2. In HIV-1-infected patients, a well-characterized exposure-response relationship was demonstrated for dolutegravir monotherapy, with levels of antiviral activity increasing with escalating doses of dolutegravir. The finding that the concentration at the end of dosing interval was the pharmacokinetic parameter that best predicted a reduction in plasma HIV-1 RNA from baseline to day 11 is consistent with the observation that antiviral activity is associated with the maintenance of therapeutic plasma concentrations throughout the dosing interval [43]. In our study, a similar behavior of dolutegravir at double daily dose could be observed on HIV-2, with highly elevated median dolutegravir C_{12h} , conferring a well-preserved inhibitory quotient value despite raltegravir-associated mutations. Moreover, in the 4 patients with available pharmacokinetic data and infected with resistant HIV-2 harboring a single integrase-associated mutation as N155H or Y143C/G/H/R, a dolutegravir C_{12h} of at least 4000 ng/mL confers an antiviral potency to the containing regimen capable of producing a $-1.5 \log_{10}$ decrease of plasma HIV-2 RNA. In another way, the higher median dolutegravir C_{12h} tended to be associated with undetectable plasma HIV-2 RNA. Finally, the good plasma exposure of 50 mg dolutegravir twice daily seems to provide advantage in terms of antiviral efficacy on resistant HIV-2, without any deleterious adverse effects.

In conclusion, these preliminary results demonstrated for the first time that dolutegravir-containing regimens might provide a substantial initial efficacy rate for salvage therapy in heavily ARV-experienced HIV-2-infected patients harboring virus without the Q148 or N155 integrase inhibitor mutation profiles. Cross-resistance of first-generation INSTIs with dolutegravir seems to be higher in HIV-2 than in HIV-1. Larger patient numbers and longer follow-up are warranted to confirm these findings.

Notes

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